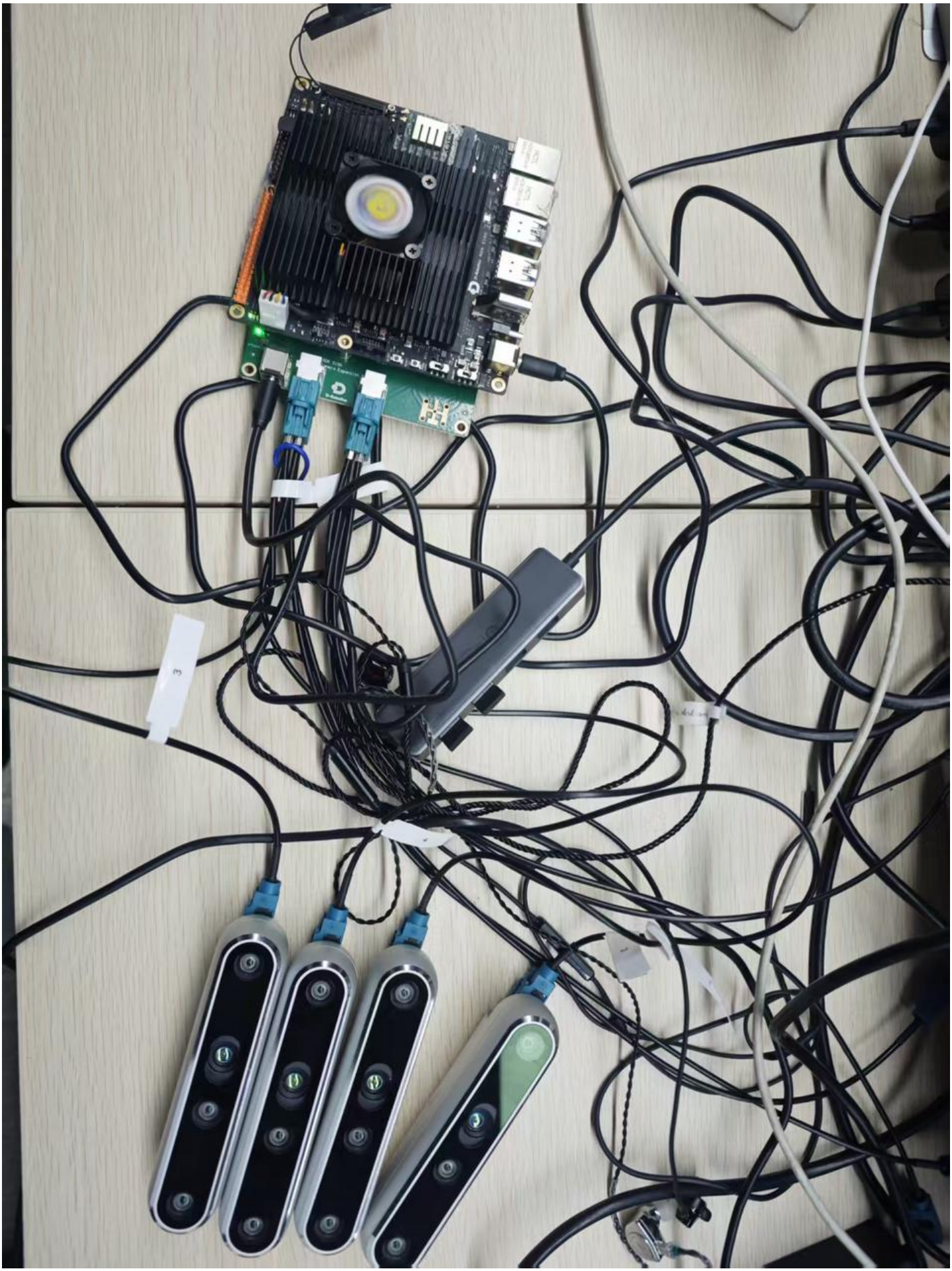
D457+ROS2驱动异常问题分析

一、问题背景

当前在使用**D457+GMSL+ROS2**长时间使用驱动过程中会出现**ROS节点重启、崩溃**的问题，该问题严重影响运动系统相关功能；

本次测试，通过GMSL以及对应cam小板连接四相机，但只启动单个相机；



二、问题记录

1. 运行过程中出现**相机断连**，重新识别相机后，ros驱动无法正常启动从而崩溃

a. 在本阶段测试中驱动运行时间分别为12min，4min，**2min（本次测试时移动了相机和线束）**
   i. 12min，3978启动，4698断连、4717尝试重启失败崩溃
   ii. 4min，549启动、558启动成功、796断连、806尝试重启失败崩溃
   iii. **2min，**378启动、383启动成功、393第一次崩溃、404重启成功、504第二次崩溃、516尝试第二次重启失败崩溃
   iv. 8min，6624启动，6632启动成功，7131崩溃、7143尝试重启失败崩溃；

代码块

```
ros2 launch realsense2_camera rs_launch.py serial_no:="\"318122302243\"" camera_name:=cam4 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-12-24-15-49-04-564124-ubuntu-46267
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [launch.user]: 🚀 Launching as Normal ROS Node
[INFO] [realsense2_camera_node-1]: process started with pid [46268]
[realsense2_camera_node-1] [INFO] [1766562549.061522175] [realsense.cam4]: RealSense ROS v4.56.4
[realsense2_camera_node-1] [INFO] [1766562549.061662650] [realsense.cam4]: Built with LibRealSense v2.56.4
[realsense2_camera_node-1] [INFO] [1766562549.061704075] [realsense.cam4]: Running with LibRealSense v2.56.5
[realsense2_camera_node-1] [WARN] [1766562549.061721200] [realsense.cam4]: ****************************************************
[realsense2_camera_node-1] [WARN] [1766562549.061732275] [realsense.cam4]: ** running with a different librealsense version **
[realsense2_camera_node-1] [WARN] [1766562549.061741800] [realsense.cam4]: ** than the one the wrapper was compiled with!   **
[realsense2_camera_node-1] [WARN] [1766562549.061751925] [realsense.cam4]: ****************************************************
[realsense2_camera_node-1] [INFO] [1766562551.416750200] [realsense.cam4]: Device with serial number 318122302243 was found.
[realsense2_camera_node-1]
[realsense2_camera_node-1] [INFO] [1766562551.416838250] [realsense.cam4]: Device with physical ID /dev/video-rs-depth-0 was found.
[realsense2_camera_node-1] [INFO] [1766562551.416867200] [realsense.cam4]: Device with name Intel RealSense D457 was found.
[realsense2_camera_node-1] [WARN] [1766562551.417136050] [realsense.cam4]: Error extracting usb port from device with physical ID: /dev/video-rs-depth-0
[realsense2_camera_node-1] Please report on github issue at https://github.com/IntelRealSense/realsense-ros
[realsense2_camera_node-1] [INFO] [1766562551.417236725] [realsense.cam4]: getParameters...
```

```
[realsense2_camera_node-1] [INFO] [1766562551.417639150] [realsense.cam4]: JSON file is not provided
[realsense2_camera_node-1] [INFO] [1766562551.417671450] [realsense.cam4]: Device Name: Intel RealSense D457
[realsense2_camera_node-1] [INFO] [1766562551.417685100] [realsense.cam4]: Device Serial No: 318122302243
[realsense2_camera_node-1] [INFO] [1766562551.417696000] [realsense.cam4]: Device physical port: /dev/video-rs-depth-0
[realsense2_camera_node-1] [INFO] [1766562551.417708550] [realsense.cam4]: Device FW version: 5.17.0.10
[realsense2_camera_node-1] [INFO] [1766562551.417718625] [realsense.cam4]: Device Product ID: 0xABCD
[realsense2_camera_node-1] [INFO] [1766562551.417728000] [realsense.cam4]: Sync Mode: Off
[realsense2_camera_node-1]  24/12 15:49:13,547 ERROR [281473357637856] (librealsense-exception.h:52) xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1]  24/12 15:49:13,547 ERROR [281473357637856] (rs.cpp:256) [rs2_set_option( options:0xaaab187efde0, option:Exposure, value:8500 ) Backend] xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1] [WARN] [1766562553.547693775] [realsense.cam4]: Set parameter {depth_module.exposure} failed: xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1]  24/12 15:49:14,601 ERROR [281473357637856] (librealsense-exception.h:52) xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1]  24/12 15:49:14,601 ERROR [281473357637856] (rs.cpp:256) [rs2_set_option( options:0xaaab187efde0, option:Exposure, value:8500 ) Backend] xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1] [WARN] [1766562554.601489250] [realsense.cam4]: Failed to set value to sensor: depth_module.exposure = 2147483647[1, 200000]
[realsense2_camera_node-1] xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1] [WARN] [1766562555.724389900] [realsense.cam4]: re-enable the stream for the change to take effect.
[realsense2_camera_node-1]  24/12 15:49:15,724 ERROR [281473357637856] (librealsense-exception.h:52) xioctl(VIDIOC_G_EXT_CTRLS) failed Last Error: Invalid argument
[realsense2_camera_node-1]  24/12 15:49:15,725 ERROR [281473357637856] (rs.cpp:256) [rs2_get_option( options:0xaaab187efe50, option_id:Exposure ) Backend] xioctl(VIDIOC_G_EXT_CTRLS) failed Last Error: Invalid argument
[realsense2_camera_node-1] [ERROR] [1766562555.725057025] [realsense.cam4]: An error has occurred while calling sensor for: rgb_camera.exposure:xioctl(VIDIOC_G_EXT_CTRLS) failed Last Error: Invalid argument
[realsense2_camera_node-1] [WARN] [1766562555.758675050] [realsense.cam4]: re-enable the stream for the change to take effect.
```

```
[realsense2_camera_node-1] [INFO] [1766562555.762722700] [realsense.cam4]: Stopping Sensor: Depth Module
[realsense2_camera_node-1] [INFO] [1766562555.762849825] [realsense.cam4]: Stopping Sensor: RGB Camera
[realsense2_camera_node-1] [INFO] [1766562555.819414600] [realsense.cam4]:
```

2. 启动时ROS节点直接崩溃，**常出现在节点异常崩溃后尝试手动重启节点时，本次3087启动、3098启动失败崩溃；**

代码块

```
root@ubuntu:~# ros2 launch realsense2_camera rs_launch.py serial_no:="\"318122302243\"" camera_name:=cam4 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15
[INFO] [launch]: All log files can be found below /root/.ros/log/2025-12-24-15-58-05-913091-ubuntu-48988
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [launch.user]: 🚀 Launching as Normal ROS Node
[INFO] [realsense2_camera_node-1]: process started with pid [48989]
[realsense2_camera_node-1] [INFO] [1766563087.368412050] [realsense.cam4]: RealSense ROS v4.56.4
[realsense2_camera_node-1] [INFO] [1766563087.368552425] [realsense.cam4]: Built with LibRealSense v2.56.4
[realsense2_camera_node-1] [INFO] [1766563087.368643025] [realsense.cam4]: Running with LibRealSense v2.56.5
[realsense2_camera_node-1] [WARN] [1766563087.368673425] [realsense.cam4]: ****************************************************
[realsense2_camera_node-1] [WARN] [1766563087.368686025] [realsense.cam4]: ** running with a different librealsense version **
[realsense2_camera_node-1] [WARN] [1766563087.368696325] [realsense.cam4]: ** than the one the wrapper was compiled with!   **
[realsense2_camera_node-1] [WARN] [1766563087.368707175] [realsense.cam4]: ****************************************************
[realsense2_camera_node-1] [INFO] [1766563090.616152125] [realsense.cam4]: Device with serial number 318122302243 was found.
[realsense2_camera_node-1]
[realsense2_camera_node-1] [INFO] [1766563090.616231375] [realsense.cam4]: Device with physical ID /dev/video-rs-depth-0 was found.
[realsense2_camera_node-1] [INFO] [1766563090.616254275] [realsense.cam4]: Device with name Intel RealSense D457 was found.
[realsense2_camera_node-1] [WARN] [1766563090.616521675] [realsense.cam4]: Error extracting usb port from device with physical ID: /dev/video-rs-depth-0
[realsense2_camera_node-1] Please report on github issue at https://github.com/IntelRealSense/realsense-ros
[realsense2_camera_node-1] [INFO] [1766563090.616664325] [realsense.cam4]: getParameters...
```

```
[realsense2_camera_node-1] [INFO] [1766563090.617082850] [realsense.cam4]: JSON file is not provided
[realsense2_camera_node-1] [INFO] [1766563090.617122675] [realsense.cam4]: Device Name: Intel RealSense D457
[realsense2_camera_node-1] [INFO] [1766563090.617138700] [realsense.cam4]: Device Serial No: 318122302243
[realsense2_camera_node-1] [INFO] [1766563090.617150100] [realsense.cam4]: Device physical port: /dev/video-rs-depth-0
[realsense2_camera_node-1] [INFO] [1766563090.617161225] [realsense.cam4]: Device FW version: 5.17.0.10
[realsense2_camera_node-1] [INFO] [1766563090.617171250] [realsense.cam4]: Device Product ID: 0xABCD
[realsense2_camera_node-1] [INFO] [1766563090.617181350] [realsense.cam4]: Sync Mode: Off
[realsense2_camera_node-1]  24/12 15:58:11,726 ERROR [281473214965984] (librealsense-exception.h:52) xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1]  24/12 15:58:11,726 ERROR [281473214965984] (rs.cpp:256) [rs2_set_option( options:0xaaab11c4fd70, option:Exposure, value:8500 ) Backend] xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1] [WARN] [1766563091.726568600] [realsense.cam4]: Set parameter {depth_module.exposure} failed: xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1]  24/12 15:58:13,772 ERROR [281473214965984] (librealsense-exception.h:52) xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1]  24/12 15:58:13,773 ERROR [281473214965984] (rs.cpp:256) [rs2_set_option( options:0xaaab11c4fd70, option:Exposure, value:8500 ) Backend] xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1] [WARN] [1766563093.773236075] [realsense.cam4]: Failed to set value to sensor: depth_module.exposure = 2147483647[1, 200000]
[realsense2_camera_node-1] xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1] [WARN] [1766563095.930409150] [realsense.cam4]: re-enable the stream for the change to take effect.
[realsense2_camera_node-1]  24/12 15:58:15,930 ERROR [281473214965984] (librealsense-exception.h:52) xioctl(VIDIOC_G_EXT_CTRLS) failed Last Error: Invalid argument
[realsense2_camera_node-1]  24/12 15:58:15,930 ERROR [281473214965984] (rs.cpp:256) [rs2_get_option( options:0xaaab11c4fde0, option_id:Exposure ) Backend] xioctl(VIDIOC_G_EXT_CTRLS) failed Last Error: Invalid argument
[realsense2_camera_node-1] [ERROR] [1766563095.930996875] [realsense.cam4]: An error has occurred while calling sensor for: rgb_camera.exposure:xioctl(VIDIOC_G_EXT_CTRLS) failed Last Error: Invalid argument
[realsense2_camera_node-1] [WARN] [1766563095.948238775] [realsense.cam4]: re-enable the stream for the change to take effect.
```

```
[realsense2_camera_node-1] [INFO] [1766563095.952801450] [realsense.cam4]: Stopping Sensor: Depth Module
[realsense2_camera_node-1] [INFO] [1766563095.952939450] [realsense.cam4]: Stopping Sensor: RGB Camera
```

```
[realsense2_camera_node-1]  24/12 15:58:11,726 ERROR [281473214965984] (librealsense-exception.h:52) xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1]  24/12 15:58:11,726 ERROR [281473214965984] (rs.cpp:256) [rs2_set_option( options:0xaaab11c4fd70, option:Exposure, value:8500 ) Backend] xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1] [WARN] [1766563091.726568600] [realsense.cam4]: Set parameter {depth_module.exposure} failed: xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1]  24/12 15:58:13,772 ERROR [281473214965984] (librealsense-exception.h:52) xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1]  24/12 15:58:13,773 ERROR [281473214965984] (rs.cpp:256) [rs2_set_option( options:0xaaab11c4fd70, option:Exposure, value:8500 ) Backend] xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1] [WARN] [1766563093.773236075] [realsense.cam4]: Failed to set value to sensor: depth_module.exposure = 2147483647[1, 200000]
[realsense2_camera_node-1] xioctl(VIDIOC_S_EXT_CTRLS) failed Last Error: Bad address
[realsense2_camera_node-1] [WARN] [1766563095.930409150] [realsense.cam4]: re-enable the stream for the change to take effect.
[realsense2_camera_node-1]  24/12 15:58:15,930 ERROR [281473214965984] (librealsense-exception.h:52) xioctl(VIDIOC_G_EXT_CTRLS) failed Last Error: Invalid argument
[realsense2_camera_node-1]  24/12 15:58:15,930 ERROR [281473214965984] (rs.cpp:256) [rs2_get_option( options:0xaaab11c4fde0, option_id:Exposure ) Backend] xioctl(VIDIOC_G_EXT_CTRLS) failed Last Error: Invalid argument
[realsense2_camera_node-1] [ERROR] [1766563095.930996875] [realsense.cam4]: An error has occurred while calling sensor for: rgb_camera.exposure:xioctl(VIDIOC_G_EXT_CTRLS) failed Last Error: Invalid argument
[realsense2_camera_node-1] [WARN] [1766563095.948238775] [realsense.cam4]: re-enable the stream for the change to take effect.
[realsense2_camera_node-1] [INFO] [1766563095.952801450] [realsense.cam4]: Stopping Sensor: Depth Module
[realsense2_camera_node-1] [INFO] [1766563095.952939450] [realsense.cam4]: Stopping Sensor: RGB Camera
[realsense2_camera_node-1] [INFO] [1766563096.019134450] [realsense.cam4]: Starting Sensor: Depth Module
[realsense2_camera_node-1] [INFO] [1766563097.155798000] [realsense.cam4]: Open profile: stream_type: Depth(0), Format: Z16, Width: 1280, Height: 720, FPS: 15
[realsense2_camera_node-1] [WARN] [1766563097.162612950] [realsense.cam4]: Frame metadata isn't available! (frame_timestamp_domain = RS2_TIMESTAMP_DOMAIN_SYSTEM_TIME)
[realsense2_camera_node-1] [INFO] [1766563097.198112075] [realsense.cam4]: Starting Sensor: RGB Camera
[realsense2_camera_node-1] [INFO] [1766563098.220010150] [realsense.cam4]: Open profile: stream_type: Color(0), Format: RGB8, Width: 1280, Height: 720, FPS: 15
[ERROR] [realsense2_camera_node-1]: process has died [pid 48989, exit code -11, cmd '/opt/ros/humble/lib/realsense2_camera/realsense2_camera_node --ros-args --log-level info --ros-args -r __node:=cam4 -r __ns:=/realsense --params-file /tmp/launch_params_0j1zuzi7 --params-file /tmp/launch_params_sh77sdvd'].
```

三、可能存在的造成问题原因

1. 结合二中的现象，所有问题均出现在相机断连之后（即出现：**[realsense2_camera_node-1] [ERROR] [1766562796.550430800] [realsense.cam4]: The device has been disconnected!**），且在今天测试过程中发现在轻轻掰gmsl接口或将相机拿起时会更容易出现断连，所以判断当前线束的可靠性需硬件侧确认；
2. ROS2驱动自身对于当前地瓜适配的方式存在缺陷，配合将现象即相关log给到GPT，分析原因

代码块

```
LibRealSense版本不一致
编译版本与运行版本不同，导致 API 调用不兼容。
GMSL接口驱动兼容性不足
RealSense D457 原生是 MIPI/USB 接口，使用 GMSL 转接时，V4L2 控制层可能不完整，导致 ioctl 调用失败。
曝光参数传递错误
出现异常的大数值（溢出/内存错误），说明驱动在参数映射时有 bug。
流重启逻辑未能恢复
错误堆叠后触发 segmentation fault，最终导致 ROS2 节点崩溃。
```